

reaching around Mach 2.9 before the motor burned out just over a minute later. The spaceship's momentum carried it onward, gradually decelerating beyond the atmosphere, before gravity took over and pulled it back toward the Earth. The "shuttlecock" effect ensured a safe re-entry, followed by a leisurely glide down to its landing site in the Mojave Desert. The goal of designing a reusable private space vehicle had been achieved. Successfully completing repeat flights on September 29, and October 4, 2004, Rutan duly claimed the prize.

### Rush into space

SpaceShipOne was only an experimental space vehicle, but its success led to immediate plans for commercial exploitation. British entrepreneur Sir

#### SPACESHIPONE

*The first privately financed reusable space vehicle, Burt Rutan's SpaceShipOne glides down to land in the Mojave Desert. SS1 alters its configuration for different stages of its flight.*



#### WHITE KNIGHT LAUNCHER

*The White Knight launcher carries SpaceShipOne under its fuselage. With two afterburning jet engines and a wingspan of 82 ft (25 m), it lifts SS1 to an altitude of 50,000 ft (15,250 m).*

Richard Branson made a deal with Rutan to set up Virgin Galactic as the first practical space tourism company. The plan was to offer suborbital flights in SpaceShipOne derivatives for around \$200,000. This would not exactly democratize space travel, but it compared favourably with the reputed \$20 million paid by Tito for his trip. Little preparation would be required beyond medical checks and the passengers would not need to wear spacesuits. Admittedly, the experience of weightlessness at the edge of space would last only a few minutes, but no one believed the flights would be short of customers. Virgin Galactic was offering the greatest fairground ride in the world.

Disappointingly, a decade after the Ansari X-Prize Virgin Galactic and its competitors were still facing insurmountable problems with safety, underlined by the fatal crash of SpaceShipTwo, the intended commercial successor to SpaceShipOne, on a test flight in October 2014. More broadly, however, the privatization of spaceflight continued to expand rapidly. Commercial companies such as SpaceX and Arianespace not only performed much of the routine work of spaceflight, such as launch of satellites and ferrying cargo to the ISS, but also pioneered new technology such as vertical-landing reusable rockets. Private space projects in various stages of development ranged from plans for orbiting space hotels to tourism on the Moon and even the founding of colonies on Mars. NASA's development of manned space flight had become

hesitant because of the difficulty of explaining its purpose to the tax-paying public. Entrepreneurs did not need to justify putting resources into space travel, if they thought customers might be prepared to pay what it cost.



#### CANADIAN ARROW

*The Canadian Arrow, an X-Prize entrant, was a conventional two-stage ground-launched rocket. Passengers would have splashed down in the ocean at the end of their trip.*

### BURT RUTAN

Born in Oregon in 1943 and raised in California, Burt Rutan is one of America's most original aircraft designers. Rutan was a flight-test project engineer for the US Air Force before setting up his own business in the 1970s. He first made his name as a designer of homebuilts, especially the innovative canard-configuration Vari-eze. In 1982 he set up his Scaled Composites company in the Mojave Desert and, four years later, earned worldwide renown with Voyager, the first

aircraft to fly nonstop around the globe without refueling. Rutan had more than 20 experimental designs to his credit before embarking on the SpaceShipOne project with which he won the Ansari X-Prize in 2004. He also found time to create GlobalFlyer, in which pilot Steve Fossett made the longest nonstop flight in history, 25,766 miles (41,467km), in 2006.

#### RUTAN IN 2005

*Burt Rutan speaking at the Smithsonian's National Air and Space Museum.*

